

LECTURE 16 TIME SERIES TOPICS

Definitions

Often econometricians have to deal with data sets that come in the form of a *time series* or *stochastic process*, a collection of observations on the same variable (or vector of variables) indexed by the date of measurement of each observation. The data are usually collected at equally spaced dates (daily, weekly, monthly, etc.) and indexed by $t = 1, \dots, T$:

$$\{Y_t : t = 1, \dots, T\}.$$

The index t measures the passage of time, and Y_t measures the evolution of the process in time. It is usually assumed that the observed sample is only a segment of the process that started in the infinite past and will go on indefinitely:

$$\{\dots, Y_{-1}, Y_0, Y_1, Y_2, \dots, Y_T, Y_{T+1}, \dots\}.$$

In the iid or *cross-sectional* case, we can view the sample as a collection of n independent “copies” of the same random variable/vector. Each Y_i samples from the common probability space, and, therefore, as the sample size increases, sample average $n^{-1} \sum_{i=1}^n Y_i$ reveals the population average $E[Y_i]$, which is the same for all i 's. The time-series case differs from the iid framework in the sense that each element in the trajectory $Y_1, Y_2, \dots, Y_T$ is observed only once. If we are interested in learning about the underlying population model, first, we should be concerned whether different observations share the same population model. It is possible that $E[Y_t] = \mu_t$ is different across t 's. The second concern is whether the measurements made at different dates effectively sample from the population model. This might not be the case if observations are highly correlated. To address these issues, we introduce the following definitions.

Definition 1. *The process Y_t is said to be strictly stationary if $(Y_{t_1}, \dots, Y_{t_k}) =^d (Y_{t_1+h}, \dots, Y_{t_k+h})$ for all k, h , and $t_1, \dots, t_k$.*

Definition 2. *Suppose that $\text{Var}(Y_t) < \infty$ for all t . The process Y_t is said to be (weakly or covariance) stationary if*

$$\begin{aligned} E[Y_t] &= \mu \text{ for all } t, \\ \text{Cov}(Y_t, Y_{t-k}) &= \text{Cov}(Y_s, Y_{s-k}) \\ &= \gamma(k) \text{ for all } s \text{ and } k. \end{aligned}$$

The function $\gamma(k)$ is called the *autocovariance function*. The stationarity assumption implies that the covariance $\text{Cov}(Y_t, Y_s) = \gamma(t-s)$, and therefore, in the scalar case,

$$\begin{aligned} \gamma(k) &= \text{Cov}(Y_t, Y_{t-k}) \\ &= \text{Cov}(Y_{t-k}, Y_t) \\ &= \gamma(-k). \end{aligned}$$

Similarly, in the vector case, we obtain that

$$\gamma(k) = \gamma(-k)^\top.$$

For a stationary process, $\text{Var}(Y_t) = \gamma(0)$. The *autocorrelation* function is defined as $\rho(k) = \gamma(k)/\gamma(0)$.

Definition 3. (loose). *The process Y_t is said to be ergodic if $\gamma(k) \rightarrow 0$ as $k \rightarrow \infty$.*

The stationarity assumptions ensure that the observations measured at different time periods share some common underlying model. The ergodicity assumption implies that distant observations are almost uncorrelated.

LLNs for a covariance stationary process

The LLN for a *strictly stationary* and *ergodic* sequence $\{Y_t : t \geq 1\}$ says that if $E[Y_t] = \mu$ (the mean is finite), then, as $T \rightarrow \infty$,

$$T^{-1} \sum_{t=1}^T Y_t \rightarrow_p \mu.$$

We will prove a weaker result. Let $\{Y_t : t \geq 1\}$ be a *covariance stationary* sequence of random variables, and assume that $\sum_{k=1}^{\infty} |\gamma(k)| < \infty$. Then, as $T \rightarrow \infty$,

$$T^{-1} \sum_{t=1}^T Y_t \rightarrow_p E[Y_t].$$

The condition $\sum_{k=1}^{\infty} |\gamma(k)| < \infty$ above says that the autocovariance function of Y_t has to be absolutely summable. For this to hold, it must be true that $\gamma(k) \rightarrow 0$ as $k \rightarrow \infty$ (Y_t is ergodic). However, contrary to the first result, here we require existence of the second moment.

Proof. First, by Markov's inequality,

$$\Pr \left(\left| T^{-1} \sum_{t=1}^T Y_t - E[Y_t] \right| > \varepsilon \right) \leq \frac{E \left| T^{-1} \sum_{t=1}^T Y_t - E[Y_t] \right|^2}{\varepsilon^2}.$$

We need to show that the numerator on the right-hand side of the above expression converges to zero as $T \rightarrow \infty$.

$$\begin{aligned} E \left| T^{-1} \sum_{t=1}^T Y_t - E[Y_t] \right|^2 &= E \left| T^{-1} \sum_{t=1}^T (Y_t - E[Y_t]) \right|^2 \\ &= T^{-2} \sum_{t=1}^T \sum_{s=1}^T (Y_t - E[Y_t]) (Y_s - E[Y_s]) \\ &= T^{-2} \sum_{t=1}^T \sum_{s=1}^T \gamma(t-s) \\ &= T^{-2} \left(T\gamma(0) + 2 \sum_{j=1}^{T-1} (T-j) \gamma(j) \right). \end{aligned}$$

To verify the last equality, observe that $\sum_{t=1}^T \sum_{s=1}^T \gamma(t-s)$ is given by the sum of all elements in the following $T \times T$ matrix.

$$\begin{pmatrix} \gamma(0) & \gamma(1) & \dots & \gamma(T-2) & \gamma(T-1) \\ \gamma(1) & \gamma(0) & \dots & \gamma(T-3) & \gamma(T-2) \\ \dots & \dots & \dots & \dots & \dots \\ \gamma(T-2) & \gamma(T-3) & \dots & \gamma(0) & \gamma(1) \\ \gamma(T-1) & \gamma(T-2) & \dots & \gamma(1) & \gamma(0) \end{pmatrix}$$

Thus, since $(T - j) / T < 1$ for $1 \leq j \leq T$, we have that

$$\begin{aligned}
\mathbb{E} \left| T^{-1} \sum_{t=1}^T Y_t - \mathbb{E}[Y_t] \right|^2 &\leq T^{-1} \left(\gamma(0) + 2 \sum_{j=1}^{T-1} \frac{T-j}{T} \gamma(j) \right) \\
&\leq T^{-1} \left(\gamma(0) + 2 \sum_{j=1}^{T-1} \frac{T-j}{T} |\gamma(j)| \right) \\
&\leq T^{-1} \left(\gamma(0) + 2 \sum_{j=1}^{T-1} |\gamma(j)| \right) \\
&\leq T^{-1} \left(\gamma(0) + 2 \sum_{j=1}^{\infty} |\gamma(j)| \right) \\
&\rightarrow 0,
\end{aligned}$$

since by the assumption $\sum_{j=1}^{\infty} |\gamma(j)| < \infty$.

The result can be extended to sequences of random p -vectors. In this case, $\gamma(j)$ is a $p \times p$ matrix. Its element (r, s) is given by $\text{Cov}(Y_{r,t}, Y_{s,t-j})$. In the vector case, we require that, for all $1 \leq r, s \leq p$, $\sum_{j=0}^{\infty} |\text{Cov}(Y_{r,t}, Y_{s,t-j})| < \infty$. In the vector case, the long-run variance-covariance matrix is given by

$$\gamma(0) + \sum_{j=1}^{\infty} \gamma(j) + \sum_{j=1}^{\infty} \gamma(j)^{\top}.$$

Examples

White noise

Suppose that

$$\begin{aligned}
\mathbb{E}[U_t] &= 0 \text{ for all } t, \\
\mathbb{E}[U_t^2] &= \sigma^2 \text{ for all } t, \\
\mathbb{E}[U_t U_s] &= 0 \text{ for all } s, t \text{ such that } s \neq t.
\end{aligned} \tag{1}$$

Such a process is called *white noise*. Sometimes condition (1) is replaced by a stronger one, which says that U_s and U_t are independent for all $s \neq t$. In this case, we say that U_t is *independent white noise*. Naturally, a white-noise process is stationary and ergodic.

Moving average models

Suppose that

$$Y_t = U_t + \theta_1 U_{t-1} + \dots + \theta_q U_{t-q},$$

where U_t is a white-noise process. Such a process is called a *moving average of order q* and denoted by $\text{MA}(q)$. We will show that a moving average process is covariance stationary. First, $\mathbb{E}[Y_t] = 0$ for all t (a nonzero mean can be obtained if we include a nonzero constant on the right-hand side of the above equation).

Next,

$$\begin{aligned}
\gamma(0) &= \sigma^2 (1 + \theta_1^2 + \dots + \theta_q^2). \\
\gamma(1) &= \text{E}[Y_t Y_{t-1}] \\
&= \text{E}[(U_t + \theta_1 U_{t-1} + \dots + \theta_q U_{t-q})(U_{t-1} + \theta_1 U_{t-2} + \dots + \theta_q U_{t-1-q})] \\
&= \theta_1 \text{E}[U_{t-1}^2] + \theta_2 \theta_1 \text{E}[U_{t-2}^2] + \dots + \theta_q \theta_{q-1} \text{E}[U_{t-q}^2] \\
&= \sigma^2 (\theta_1 + \theta_2 \theta_1 + \dots + \theta_q \theta_{q-1}). \\
&\dots \\
\gamma(q) &= \sigma^2 \theta_q. \\
\gamma(k) &= 0 \text{ for } k > q.
\end{aligned}$$

Hence, MA(q) is covariance stationary and ergodic.

Autoregressive models

Suppose that

$$Y_t = \beta_1 Y_{t-1} + \dots + \beta_p Y_{t-p} + U_t,$$

where U_t is a white-noise process. Such a process is called an autoregression of order p and denoted by AR(p). Nonzero-mean processes can be modelled by including an intercept in the above equation.

Consider the case of AR(1). Write

$$\begin{aligned}
Y_t &= \beta Y_{t-1} + U_t \\
&= \beta^2 Y_{t-2} + \beta U_{t-1} + U_t \\
&= \beta^t Y_0 + \sum_{j=0}^{t-1} \beta^j U_{t-j}.
\end{aligned}$$

Assume that $|\beta| < 1$. Then, $\lim_{j \rightarrow \infty} \beta^j = 0$, and, therefore,

$$Y_t = \sum_{j=0}^{\infty} \beta^j U_{t-j}.$$

This is called the MA(∞) representation of an AR(1) process. We have

$$\begin{aligned}
\gamma(0) &= \sigma^2 \sum_{j=0}^{\infty} \beta^{2j} \\
&= \frac{\sigma^2}{1 - \beta^2}, \\
\gamma(1) &= \sigma^2 \sum_{j=0}^{\infty} \beta^j \beta^{j+1} \\
&= \beta \sigma^2 \sum_{j=0}^{\infty} \beta^{2j} \\
&= \frac{\beta \sigma^2}{1 - \beta^2}. \\
\gamma(k) &= \sigma^2 \sum_{j=0}^{\infty} \beta^j \beta^{j+k} \\
&= \frac{\beta^k \sigma^2}{1 - \beta^2}.
\end{aligned}$$

We have that the sequence of autocovariances $\gamma(j)$ is independent of t , and, therefore, AR(1) is weakly stationary. Further, it is ergodic since $\lim_{j \rightarrow \infty} \gamma(j) = 0$. The long-run variance of the AR(1) process is given by

$$\begin{aligned} & \frac{\sigma^2}{1 - \beta^2} + 2 \sum_{j=1}^{\infty} \frac{\beta^j \sigma^2}{1 - \beta^2} \\ &= \frac{\sigma^2}{1 - \beta^2} \left(1 + 2\beta \sum_{j=0}^{\infty} \beta^j \right) \\ &= \frac{\sigma^2}{1 - \beta^2} \left(1 + 2 \frac{\beta}{1 - \beta} \right) \\ &= \frac{\sigma^2}{(1 - \beta)^2}. \end{aligned}$$

Wold decomposition

The MA(∞) representation plays an important role in time-series analysis due to the result called the Wold decomposition. According to it, if $\{Y_t\}$ is a covariance stationary and ergodic process, then there exist a white-noise sequence $\{U_t\}$ and a sequence of constants $\{\theta_j\}$ such that

$$\begin{aligned} Y_t &= \mu + \sum_{j=0}^{\infty} \theta_j U_{t-j}, \text{ where} \\ \sum_{j=0}^{\infty} \theta_j^2 &< \infty. \end{aligned}$$

Such a process is called *linear*.

CLT for linear processes

Let Y_t be a linear process:

$$Y_t = \sum_{j=0}^{\infty} \theta_j U_{t-j},$$

where $\{U_t\}$ are iid with $E[U_t] = 0$ and $E[U_t^2] = \sigma^2 < \infty$, the sequence $\{\theta_j\}$ is absolutely summable, and $\sum_{j=0}^{\infty} \theta_j \neq 0$. Then,

$$T^{-1/2} \sum_{t=1}^T Y_t \rightarrow_d N \left(0, \sum_{j=-\infty}^{\infty} \gamma(j) \right),$$

where $\gamma(j)$ is the j -th autocovariance of Y_t , and, therefore, $\sum_{j=-\infty}^{\infty} \gamma(j)$ is the long-run variance of Y_t . For example, in the scalar case, the long-run variance is given by

$$\sigma^2 \left(\sum_{j=0}^{\infty} \theta_j \right)^2.$$

Time-series regression

Suppose that the econometrician observes $\{(Y_t, X_t) : t = 1, \dots, T\}$, where

- $Y_t = X_t^\top \beta + U_t$.
- $\beta \in R^k$ is a vector of unknown regression coefficients.
- $\{X_t\}$ is strictly stationary and ergodic with finite second moments, and $E[X_t X_t^\top]$ is positive definite.
- $\{U_t X_t\}$ is a vector process and satisfies the conditions of the CLT for linear processes.

Consider the OLS estimator of β :

$$T^{1/2} (\hat{\beta}_T - \beta) = \left(T^{-1} \sum_{t=1}^T X_t X_t^\top \right)^{-1} T^{-1/2} \sum_{t=1}^T U_t X_t.$$

By the LLN, we have that

$$\left(T^{-1} \sum_{t=1}^T X_t X_t^\top \right)^{-1} \rightarrow_p (E[X_t X_t^\top])^{-1}.$$

The CLT implies that

$$T^{-1/2} \sum_{t=1}^T U_t X_t \rightarrow_d N(0, \Omega),$$

where Ω is the long-run variance-covariance matrix of $U_t X_t$:

$$\Omega = E[U_t^2 X_t X_t^\top] + \sum_{j=1}^{\infty} E[U_t U_{t-j} (X_t X_{t-j}^\top + X_{t-j} X_t^\top)].$$

Therefore,

$$T^{1/2} (\hat{\beta}_T - \beta) \rightarrow_d N(0, V),$$

where

$$V = (E[X_t X_t^\top])^{-1} \Omega (E[X_t X_t^\top])^{-1}.$$

In this case, we allow U_t to be heteroskedastic and correlated across t (serially correlated). The result is similar to the iid case, except for the expression for Ω . The asymptotic variance matrix V can be consistently estimated by

$$\hat{V}_T = \left(T^{-1} \sum_{t=1}^T X_t X_t^\top \right)^{-1} \hat{\Omega}_T \left(T^{-1} \sum_{t=1}^T X_t X_t^\top \right)^{-1},$$

where $\hat{\Omega}_T$ is the heteroskedasticity and autocorrelation consistent (HAC) estimator of Ω of Newey and West (1987, Econometrica):

$$\hat{\Omega}_T = T^{-1} \sum_{t=1}^T \hat{U}_t^2 X_t X_t^\top + T^{-1} \sum_{l=1}^L \sum_{t=l+1}^T \left(1 - \frac{l}{L+1} \right) \hat{U}_t \hat{U}_{t-l} (X_t X_{t-l}^\top + X_{t-l} X_t^\top),$$

where L is called the truncation parameter. The HAC estimator is consistent if $L \rightarrow \infty$ at a suitable rate (slower than T).